

CHAPTER 6

ADAPTATIONS TO AEROBIC TRAINING

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After completing this chapter, you will be able to

- identify and describe the impact of chronic aerobic endurance training on the physiological characteristics of the cardiovascular, respiratory, nervous, muscular, bone and connective tissue, and endocrine systems;
- recognize the interaction between aerobic endurance training and optimizing physiological responses of all body systems;
- identify and describe external factors that influence adaptations to chronic aerobic exercise including altitude, heat, smoking and vaping, blood doping, and detraining; and
- recognize the physiological differences between altitude training and heat acclimation.

Understanding how body systems adapt to long-term aerobic exercise is essential for strength and conditioning professionals to design effective training programs. This chapter explores the physiological changes that occur in response to sustained aerobic

endurance training. It also examines how external factors—including altitude, heat, detraining, and blood doping—affect the body's responses and adaptations to this type of exercise. By covering both the fundamental adaptations and the impact of these external influences, the chapter provides a comprehensive overview of aerobic endurance training effects.

Chronic Adaptations to Aerobic Exercise

Understanding the effects of aerobic endurance training on the body systems is important for assessing physical or athletic performance and determining the impact of training programs. This section deals with the effects of aerobic endurance training on the cardiovascular, respiratory, nervous, muscular, bone and connective tissue, and endocrine systems of the body ([table 6.1](#)).

Cardiovascular Adaptations

Aerobic endurance training results in several changes in cardiovascular function, including increased maximal cardiac output, increased stroke volume, and reduced heart rate at rest and during submaximal exercise. In addition, muscle fiber capillary density increases as a result of aerobic endurance training, supporting delivery of oxygen and removal of carbon dioxide.

TABLE 6.1 Physiological Adaptations to Aerobic Endurance Training

Variable	Aerobic endurance training adaptations
PERFORMANCE	
Muscular strength	No change
Muscular endurance	Increases for low power output
Aerobic power	Increases
Exercise economy	Increases
Maximal rate of force production	No change or decreases
Anaerobic power	No change
Sprint speed	No change
CARDIOVASCULAR FUNCTION	
<i>Heart rate</i>	

Note: ATP = adenosine triphosphate; ATPase = adenosine triphosphatase.

Variable	Aerobic endurance training adaptations
Resting and submaximal exercise	Decreases
Maximal exercise	No change or decreases
Stroke volume	Increases
<i>Cardiac output</i>	
Resting or submaximal exercise	No change or decreases
Maximal exercise	Increases

MUSCLE FIBERS

Fiber size	No change or increases slightly
Capillary density	Increases
Mitochondrial density	Increases
Cytoplasmic density	No change
Myosin heavy chain protein	No change or decreases in amount
<i>Myofibrillar</i>	
Packing density	No change
Volume	No change

ENZYME ACTIVITY

Creatine phosphokinase	Increases
Myokinase	Increases
Phosphofructokinase	Variable
Lactate dehydrogenase	Variable
Sodium–potassium ATPase	May slightly increase

METABOLIC ENERGY STORES

Stored ATP	Increases
Stored creatine phosphate	Increases
Stored glycogen	Increases

Note: ATP = adenosine triphosphate; ATPase = adenosine triphosphatase.

Variable	Aerobic endurance training adaptations
Stored triglycerides	Increase
Fat oxidation	Increases

CONNECTIVE TISSUE

Ligament strength	Increases
Tendon strength	Increases
Collagen content	Variable
Bone density	No change or increases

BODY COMPOSITION

% body fat	Decreases
Fat-free mass	No change

Note: ATP = adenosine triphosphate; ATPase = adenosine triphosphatase.

The most significant change in cardiovascular function with long-term (6-12 months) aerobic endurance training is the increase in maximal cardiac output, resulting primarily from improved stroke volume; maximal heart rate generally remains unaltered or may even decrease slightly. Because maximal heart rate may remain unaltered or actually decrease slightly with prolonged training (95), perhaps as a result of increased parasympathetic tone (13), the size of the left ventricle (both chamber volume and wall thickness) and the strength of contractions (increased contractility) are the key to increasing stroke volume with maximal as well as submaximal exercise. Consequently, a significantly lower heart rate in response to a standardized submaximal level of work is another hallmark of aerobic endurance training. Furthermore, heart rate increases more slowly in trained athletes than in sedentary people for a given workload (38, 108).

Another notable change that comes with aerobic endurance training is the peripheral cardiovascular adaptations at the level of the arterioles and capillaries within the muscles (108). Increased

muscle fiber capillary density has been observed in response to aerobic endurance training and is a function of the volume and intensity of training. In addition to delivering oxygen, nutrients, and hormones, the capillary circulation provides the means for removing heat and metabolic by-products. Subsequently, increasing capillary density decreases the diffusion distance for oxygen and metabolic substrates, enabling an improved exchange during subsequent aerobic endurance training and performances (88).

The implication of the abovementioned changes is that they increase an aerobic endurance athlete's ability to improve his or her maximal oxygen uptake, which is of paramount importance for optimal aerobic endurance performance. The reason for this is that a close relationship exists between maximal cardiac output and maximal oxygen uptake, the latter being the product of maximum cardiac output and arteriovenous oxygen difference (108). More simply, an increase in maximal oxygen uptake means more oxygen is delivered to and extracted by the working muscles, thereby providing an increase in exercise intensity and ultimately performance.

While not related to aerobic endurance performance outcomes, it is also worth noting that a common cardiovascular adaptation following aerobic endurance training is a lower resting heart rate. The normal discharge rate of the sinoatrial (SA) node ranges from 60 to 80 times per minute. Aerobic endurance training results in a significantly slower discharge rate due to an increase in parasympathetic tone. Increased stroke volume also affects the resting heart rate, because more blood is pumped per contraction, so the heart needs to contract less frequently to meet the same cardiac output. Aerobic endurance training can increase the heart's ability to pump blood per contraction at rest and thus may account for some of the significant **bradycardia** (slower heart rate) observed in highly conditioned aerobic endurance athletes, whose resting heart rates commonly range from 40 to 60 beats/min (58, 108). In addition to having low resting heart rates, aerobic

endurance-trained athletes have also been found to have an accelerated heart rate recovery after exercise (13). This is important to know for athlete training and monitoring purposes.

Respiratory Adaptations

Pulmonary ventilation generally does not limit aerobic exercise and is either unaffected or only moderately affected by training (8, 32). While the respiratory system can accommodate the demands of high-intensity aerobic endurance exercise for the majority of athletes, evidence suggests that in some highly trained aerobic endurance athletes, such total body aerobic endurance exercise may exceed the respiratory system's functional capacity and affect the ability to transport oxygen to locomotor musculature (2). Furthermore, the ventilatory adaptations observed appear to be highly specific to activities that involve the type of exercise used in training; that is, adaptations observed during lower extremity exercise primarily occur as a result of lower extremity training (108). If exercise training focuses on the lower extremities, thereby reducing lactate levels and carbon dioxide production from lactate buffering as training adaptations take place, one will not likely observe ventilatory adaptation during upper extremity activities, where no training adaptations have taken place (108). Training adaptations include increased tidal volume and breathing frequency with maximal exercise. With submaximal activity, breathing frequency is often reduced and tidal volume is increased. Ventilatory adaptations result from local, neural, or chemical adaptations in the specific muscles trained through exercise (8, 32).

Neural Adaptations

While neural adaptations resulting from strength and power training and different speeds of locomotion have been thoroughly investigated (134), as well as those from fatiguing static and dynamic contraction perspectives (152), those arising from chronic aerobic training remain relatively understudied. Nonetheless, chronic

aerobic training in disciplines such as cycling and running can lead to beneficial adaptations in movement economy through improvements in motor recruitment. Chronic aerobic training can result in increased muscle coactivation, enhanced leg stiffness, and a greater ratio of eccentric-to-concentric muscle activity, enabling athletes to use stored elastic energy more efficiently, reducing the metabolic cost of exercise (73a). Supporting research has shown that trained cyclists and runners exhibit less variability in muscle activity between pedal strokes and strides, respectively, as well as shorter durations of muscle activity compared to their novice counterparts (21a). Furthermore, multisport athletes, such as triathletes, demonstrate more extensive and variable muscle coactivation, along with less modulation and refined patterns of muscle activity during cycling and running when compared to single-sport athletes (21a, 21b) reflecting interference with neuromuscular adaptations from multisport training.

Muscular Adaptations

One of the fundamental adaptive responses to aerobic endurance training is an increase in the aerobic capacity of the trained musculature. This adaptation allows the athlete to perform a given absolute intensity of exercise with greater ease. More impressively, after training, an athlete can exercise at a greater relative intensity of a now-higher maximal aerobic power. Thus, measuring an athlete's maximal oxygen uptake only before and after aerobic endurance training may not accurately portray his or her ability to perform during competition. For example, an athlete who can run the marathon at a pace equal to 75% of maximal oxygen uptake may, after training, be able to maintain a pace that is 80% of maximal aerobic power. This adaptation occurs as a result of an improvement in what is called the athlete's **critical power** or, more simply, the athlete's maximal oxidative energy capacity, which also means that the exercising muscle's rate of lactate production matches the rate of clearance by muscles and other tissues (127a).

Consequently, the **onset of blood lactate accumulation** (OBLA) (see [chapter 3](#)) occurs at a higher percentage (up to 80%-90%) of the trained athlete's aerobic capacity.

The muscular component of an aerobic endurance training program involves submaximal muscle contractions extended over a large number of repetitions with little recovery. Therefore, the relative intensity is very low, and the overall volume is very high. This manner of training encourages relative increases in aerobic potential that are similar in type I and type II fibers (167). Compared to type II fibers, type I fibers have a higher preexisting initial aerobic capacity, to which the increase in aerobic potential from training is added (53). Thus, type I fibers possess an oxidative capacity greater than that of type II fibers both before and after training. However, it should be noted while aerobic endurance training has been found to encourage relative increases in aerobic potential that are similar in type I and type II fibers (167), such training has also been found to reduce type I and type II fiber size while maintaining or improving the functional ability of the fibers (157). After 13 weeks of marathon running training, a reduction of 20% in muscle fiber size in both fiber types was found, while strength in both fiber types was maintained and actually increased by 18% in type IIa fibers following a three-week taper (157).

Little evidence shows that type II fibers change into type I fibers as a result of aerobic endurance training, but there may be a gradual conversion within the two major type II fiber subgroups of type IIX fibers to type IIa fibers (4, 167). This adaptation is significant, in that type IIa fibers possess a greater oxidative capacity than type IIX fibers and have functional characteristics more similar to those of type I fibers. However, it should also be noted that **hybrid fibers**, which can have a combination of two or three isoforms (pure fibers), are also present in skeletal muscles and can experience similar adaptations (80). Hybrid fibers can increase or decrease depending on the muscle's composition before exercise as well as the exercise stimulus encountered (110). In the case of an athlete undertaking

aerobic endurance exercise, for example, a hybrid I/IIa fiber could transform to a type I fiber (110). This scenario was highlighted by Trappe and colleagues (157), who examined the impact of aerobic endurance training on muscle fiber adaptations in runners participating in a 16-week marathon training study including a 3-week taper and marathon. The results showed approximately a 50% decline in the percentage of hybrid fibers and an increase in the percentage of type I fibers in the gastrocnemius muscle. While our understanding of how hybrid fibers in humans adapt to various training environments is still evolving, we do know that hybrid fibers in aerobic endurance runners are influenced by training volume, and the content of hybrid fibers (i.e., distribution of MHC isoforms) differs between longer- versus shorter-distance runners (80). Further, the abovementioned conversions following aerobic endurance training, whether within fiber type or from hybrid fibers, results in a more oxidative phenotype of the muscle that contributes to improved aerobic endurance performance.

At the cellular level, muscular adaptations to aerobic exercise include an increase in the size and number of mitochondria (68) and increased myoglobin content (26, 64). **Myoglobin** is a protein that transports oxygen within the cell. **Mitochondria** are the organelles in cells that are responsible for aerobically producing adenosine triphosphate (ATP) via oxidation of pyruvate and lactate (the products of glycolysis) and free fatty acids. When the larger and more numerous mitochondria are combined with an increase in the quantity of oxygen that can be delivered to mitochondria by the greater concentration of myoglobin, the capacity of the muscle tissue to extract and use oxygen is enhanced. This adaptation is further augmented by increases in the level and activity of the enzymes involved in oxidative metabolism (73) and a parallel increase in glycogen (51, 53) and triglyceride (113) stores. Also worth noting is that aerobic endurance training can spare muscle glycogen (less glycogen use during exercise) and increase fat use within the muscle, which helps prolong performance at the same exercise

intensity (71). The training induced an increase in fat use attributed to the increase in fat-metabolizing enzymes and greater mitochondrial respiratory capacity within the exercising muscles (108).

Bone and Connective Tissue Adaptations

The selection of different forms of aerobic exercise has met with some success in bringing about improvements in bone mass. Aerobic programs that are the most successful in stimulating bone growth involve more intense physical activities such as running and high-intensity aerobics (12, 17). Running promotes bone growth more effectively than other types of aerobic exercises such as cycling and swimming. This is due to the repetitive high-impact loading that occurs during running, in which the body experiences forces several times its own weight. In contrast, cycling and swimming are weight-supported activities that place minimal strain on the skeletal system, resulting in less stimulation for bone growth (150). In fact, non-weight-bearing aerobic exercise can reduce bone density, particularly in areas not used to produce locomotion. For example, in a study comparing athletes who were runners, cyclists, or both, it was reported that individuals who were runners had greater total and leg bone mineral density when compared to non-exercising control participants, while cyclists had a reduced lower spine bone mineral density than the non-exercising control group, and those in both groups had greater total and arm bone mineral density compared to non-exercising control participants (150). These findings suggest that a sport like cycling may not produce enough strain to improve bone mass not directly related to locomotion. As such, the inclusion of other weight-bearing activities is likely warranted to help counteract the potential reductions in bone mass associated with non-weight-bearing activities (150). This research highlights that skeletal adaptations brought about by aerobic endurance training are specific to the activity undertaken and, more specifically, the axial loading that the activity induces.

The key to the success of aerobic exercise in stimulating new bone formation is that the activity must be significantly more intense than the daily activities the person normally engages in so as to exceed the minimum threshold intensity, as well as at a cyclical strain to exceed the minimum and strain frequency for bone growth (12). The intensity of the activity must systematically increase in order to continually overload the bone. Eventually, it may become difficult to overload bone through aerobic exercise when the oxygen transport system rather than limitations of the musculoskeletal system restricts the progression to new exercise intensity. Bone responds to the magnitude and rate of external loading. Therefore, to enhance the stimulus to the musculoskeletal system, it is also necessary to increase the rate of limb movement. Using high-intensity interval training techniques is one method of providing a greater osteogenic stimulus while still providing the benefits associated with aerobic exercise (14, 39). Another method to increase the musculoskeletal system stimulus is to combine aerobic endurance training with resistance training, a known stimulus for improving bone mass (107), in what is commonly referred to as **concurrent training** (146) (refer to [chapter 21](#) for more information).

In mature adults, the extent to which tendons, ligaments, and cartilage grow and become stronger is proportional to the intensity of the exercise stimulus, especially from weight-bearing activities (119). Tendons and ligaments consist of bundles of collagen-containing fibers that increase in number, size, and density following consistent exercise stimulation (44). As with bone and muscle, exercise intensity that consistently exceeds the strain placed on the connective tissues during normal daily activities is needed to create connective tissue changes. An example of the positive effects of weight-bearing activity on cartilage is evidenced in a typical knee joint, where the surfaces of the joint that experience the greatest degree of weight bearing are thicker than the non-weight-bearing surfaces (119). Complete movement through a full range of motion

during weight bearing is likely essential to maintain tissue viability (148).

Aerobic exercise, particularly running, has received a lot of research attention due not only to the health benefits it can provide but also because of suggestions that habitual running can have a negative impact on joints like the knees and hips, increasing the risk of cartilage damage and osteoarthritis. While mixed or inconclusive findings on the impact of running on knee health have been reported (154), a study evaluating thousands of non-elite male and female runners across the life span (ages from 12 to over 50) reported that running did not increase the odds of knee pain or symptomatic knee osteoarthritis (97). Similarly, a study has reported that running does not increase the risk of symptomatic knee osteoarthritis compared to non-running activities (98). In individuals without preexisting osteoarthritis, running does not appear to harm the knees (98). Moreover, some research suggests that running may even offer protection against general knee pain (29). This protection may be associated with the improved periarticular strength, not to mention proprioception that comes with the running motion (98).

In studies of elite aerobic endurance-trained athletes, the results are similar to those of lesser-trained runners. For example, in a study of over 2,000 former elite athletes, it was found that those from a power and mixed-sports background (e.g., soccer, basketball, track and field) had an increased rate of hospital admissions for premature osteoarthritis compared to aerobic endurance athletes (e.g., long-distance runners), who also experienced admissions but at a notably older age (90). A similar finding was reported in another study of various former athletes including elite male aerobic endurance and track and field athletes, where it was found that aerobic endurance athletes reported fewer hip issues than control subjects (78). It has been suggested that aerobic endurance athletes may benefit from having a lower body mass index, which assists in preventing osteoarthritis, along with having a more physically active lifestyle, which may assist with a higher-than-average life

expectancy, thus explaining the later onset of osteoarthritis, if it does occur (90). While these studies did not highlight any notable differences in the outcomes between the elite and non-elite running athletes, it has been suggested that the level of exposure may be a risk factor for knee osteoarthritis (98), as a result of the duration, intensity, and frequency of training.

Endocrine Adaptations

Just as endocrine responses to resistance training are needed for muscle development (87), the changes in hormone production that contribute to the body's adaptation to aerobic exercise are equally important (45, 46, 82, 108, 113). Testosterone, insulin, insulin-like growth factors (IGF-1), and growth hormone affect the integrity of muscle, bone, and connective tissue as well as assist in maintaining metabolism within a normal range (40, 41, 84, 85, 158). The endocrine adaptations following aerobic endurance training vary according to the level of exercise intensity and duration, with some hormonal responses increasing while others decrease, as outlined in the following.

High-intensity aerobic training augments the absolute secretion rates of many hormones in response to maximal exercise, although trained athletes have attenuated hormonal responses to submaximal exercise (103). However, the hormonal response of trained athletes is similar to an untrained person when performing aerobic endurance exercise at the same relative submaximal exercise intensity (108). The greater hormonal response patterns to maximal exercise appear to augment the athlete's ability to tolerate and sustain prolonged high-intensity exercise (170). When exercise intensity is very high and exercise duration is very short (from 5-10 s), only "fight-or-flight" changes in peripheral blood hormone concentrations occur (e.g., epinephrine and norepinephrine concentrations increase) (86).

The key to endocrine responses, and hence adaptations for aerobic endurance athletes, may be the interaction between training variables such as volume and intensity plus the duration of the

activity and its influence on energy supply. Over the course of a season, high-level male aerobic endurance runners' basal concentrations of certain hormones (e.g., testosterone and insulin) may change and be more sensitive to variances in training parameters such as intensity and distance run each week than hormones such as cortisol (1). Further, these responses may also be influenced by energy availability, with basal testosterone concentrations decreasing for athletes who ran a greater number of miles and had low energy availability (1). Additionally, research on elite rowers has demonstrated that certain hormones (testosterone and cortisol) may be more sensitive to changes in training volume than hormones such as growth hormone (129). Also worth noting is that low energy availability can affect the degree of protein involvement in aerobic endurance exercise, particularly during bouts of long duration (94). Therefore, consideration must be given over the course of a season as to how the training variables are manipulated and energy supplies are affected, not to mention how an athlete recovers (i.e., type of recovery strategies employed including nutrition), because these factors will determine the endocrine response as well as the adaptations experienced during and after training. An important point related to all this information is what is known as *relative energy deficiency in sport (RED-S)*. For more detail, see [chapter 11](#).

Aerobic training, especially running, is often associated with an acute increase in net protein breakdown from the muscle (155), brought about in part by acute stress-induced cortisol secretion (142, 143) despite concomitant increases in anabolic hormones such as testosterone and IGF-1 (160). However, evidence suggests that net protein synthesis in skeletal muscle of aerobic endurance-trained athletes does occur and may lead to muscle hypertrophy (81), but it is most likely due to mitochondrial rather than contractile proteins (161, 166). Further, the turnover rate of mitochondrial proteins and myofibrillar proteins may be different, which may influence the recovery capacity of the muscle after exercise (145). This may be

due to the impact that protein ingestion has on recovery following different forms of aerobic endurance training such as high- and low-intensity sessions; some evidence suggests that the combination of these variables may alter the timing (i.e., give preference to one pathway over another) and rate of muscle protein synthesis of myofibrillar and mitochondrial proteins (145). Consequently, these protein pathways can also be influenced by exercise–nutrient interactions, especially in the early stages of recovery after exercise, hence the importance of post-exercise recovery nutrition that incorporates the ingestion of high-quality proteins in addition to carbohydrates (145).

Adaptations to Aerobic Endurance Training

Much research has been done on training adaptations associated with aerobic endurance exercise (5, 15, 20, 34, 42, 65, 66, 161). Aerobic metabolism plays a vital role in human performance and is basic to all sports, if for no other reason than recovery (156). Metabolically, the Krebs cycle and electron transport chain are the main pathways in aerobic energy production. Aerobic metabolism produces far more ATP energy than anaerobic metabolism and uses fats, carbohydrates, and (minimally) proteins as fuel sources for generating ATP. Many sports involve interactions between the aerobic and anaerobic metabolic systems and thus require appropriate training. For example, soccer, lacrosse, basketball, field hockey, and ice hockey involve continuous movement (and thus constant aerobic demand) mixed with bursts of sprint and power activities. Proper conditioning of the aerobic system is vital to the athlete's ability to sustain such activity and adequately recover within and between exercise sessions.

What Improvements in Performance Result From Aerobic Exercise?

The body systems affected by aerobic exercise include musculoskeletal, cardiovascular, and respiratory. Adaptations include the following:

- **Respiratory system.** Decreased submaximal respiration rate (37).
- **Cardiovascular system.** Decreased heart rate for fixed submaximal workloads (131) associated with increased stroke volume and cardiac output. Blood (plasma) volume is also increased, supporting increased stroke volume and cardiac output (55). *Hemoglobin* mass also increases but not the percentage of hemoglobin (i.e., hematocrit), which may decrease slightly due to the relatively greater increase in plasma volume (140).
- **Musculoskeletal system.** Increased arterial–venous O₂ difference associated with increased capillarization in muscle (136, 153), increased oxidative enzyme concentrations, and increased mitochondrial size and density (70).

- **Aerobic power (maximal oxygen uptake).** Arguably the most significant change in physiological variables with aerobic exercise training is an increase in maximal oxygen uptake ($\dot{V}O_{2\max}$), the criterion index of cardiovascular fitness. The $\dot{V}O_{2\max}$ is determined in part by genetics and by the training program undertaken by the athlete. Elite athletes may show minor changes in $\dot{V}O_{2\max}$ with training (5%-10%), while untrained individuals may increase $\dot{V}O_{2\max}$ by as much as 20% (67, 75). A high $\dot{V}O_{2\max}$ coupled with increased lactate threshold allows enhanced performance for running sports as well as sports requiring intermittent sprinting (soccer, basketball, and other team sports).

- **Lactate threshold.** Aerobic training increases the absolute lactate threshold, allowing the highly trained individual to work at both a higher relative and absolute percentage of his or her $\dot{V}O_{2\max}$ than a less-trained individual. This increased tolerance of lactate translates into a variety of performance outcomes; these include running at a higher percentage of $\dot{V}O_{2\max}$ for a race (resulting in reduced time), covering more distance during a game, enhanced recovery for second-half performance, and working at higher exercise intensities throughout an event. The importance of having a high lactate threshold can be seen in the following example. Two individuals can have the same $\dot{V}O_{2\max}$, perhaps $50 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$; however, if one has a lactate threshold of 80% of $\dot{V}O_{2\max}$ (i.e., threshold occurs at $40 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) whereas the other's occurs at 70% (i.e., $35 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$), the first individual will be able to maintain a power output equivalent to $5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ higher than the second individual. All other things being equal, such as exercise economy, this would result in a greater speed of movement and thus superior performance.

- **Effective use of substrate.** Carbohydrates are the preferred source of fuel for high-intensity intermittent exercise in many team sports. Aerobic exercise training results in a greater use of fat as a substrate for exercise with a relative sparing of carbohydrate. With a sparing of carbohydrate (i.e., reduced glycogen depletion), an aerobic endurance-trained individual can maintain higher-intensity exercise for longer periods of time. Aerobic exercise may be further improved through various carbohydrate-loading manipulations to increase endogenous glycogen stores (19).

- **Muscle fiber adaptations.** When muscle fibers are examined cross-sectionally, elite distance runners have a higher proportion (percentage) of type I fibers, and the available type I fibers are functionally very efficient for aerobic metabolism (increased mitochondrial density and oxidative enzyme capacity [69] and capillary network for oxygen delivery). Aerobic exercise training, especially long-distance and high-intensity intermittent exercise, results in an increase in the oxidative capacity of type I fibers. Research indicates that after a period of aerobic endurance training such as marathon training, there is a shift toward type I fibers associated with the higher volume of aerobic endurance training. The oxidative phenotype type I/IIa fibers also increase, while type IIa and IIa/x decrease in both vastus lateralis and soleus. However, it should be considered that certain muscles

may have a higher propensity for fiber shifting and other factors such as the tapering or other phase of training may affect these shifts (125, 102). From a performance standpoint, these metabolic and fiber changes result in a more efficient use of aerobic energy production.

- **Exercise economy.** The economy of exercise is mostly a function of biomechanics and technique, which are influenced by the athlete's metabolic, cardiovascular, and neuromuscular efficiency and training (7). Two aerobic endurance athletes may have the same maximal oxygen uptake and lactate threshold yet not show the same performance outcomes. The athlete with better economy would be able to race at a higher target pace or power for the same amount of energy consumption, likely resulting in a better performance, even though the two athletes have the same $\dot{V}O_{2\max}$ and lactate threshold (23).

Every athlete needs a basic level of aerobic endurance—if not for performance, for health reasons—that can be achieved using a wide variety of training modalities and programs (see [chapter 21](#) for more information). Apart from the stereotypical submaximal aerobic training methods, another method that can produce significant gains in aerobic fitness (e.g., increased $\dot{V}O_{2\max}$, lactate threshold) is interval training (16, 42, 92).

One of the most measured adaptations to aerobic endurance training is an increase in maximal oxygen uptake associated with an increase in maximal cardiac output (15, 31, 65, 161). As the intensity of exercise increases, oxygen consumption rises to maximal levels. When oxygen consumption can no longer increase to meet the demands, maximal oxygen uptake has been achieved even in the presence of continuing oxygen availability. Aerobic endurance training can improve an athlete's aerobic power by 5% to 30%, depending, in part, on the starting fitness level as well as the genetic potential of the individual (5). Most adaptations in maximal oxygen consumption can be achieved within a 6- to 12-month training period. After that, further changes in aerobic endurance performance consist of increases in **exercise economy** and increased lactate threshold (75). Metabolic changes include increased respiratory capacity, lower blood lactate concentrations at a given submaximal exercise intensity, increased mitochondrial and

capillary densities, and improved enzyme activity. It may be that experienced athletes do not experience further increases in their $\dot{V}O_2\text{max}$ through chronic aerobic endurance training but that their performance improves due to enhanced exercise economy (15, 66).

The intensity of training is one of the most important factors in improving and maintaining aerobic power. Short, high-intensity bouts of interval sprints can improve maximal oxygen uptake if the interim rest period is also short. A **repeated sprint-sequence interval session** can generally be defined as at least two short bouts of ≥ 10 seconds with a short recovery period of ≤ 60 seconds; however, this will depend on the athletes' goals (50). Callister and colleagues (20) demonstrated that long rest periods used with sprints improve sprint speed without significant increases in maximal aerobic power. Therefore, longer training sessions with higher amounts of rest between exercise bouts result in less improvement in aerobic capacity. The use of shorter recovery periods between high-intensity training intervals has been shown in various studies to improve various skeletal muscle metabolic processes, resulting in improved aerobic endurance performance (48).

Aerobic endurance training is generally associated with alterations in body composition, assuming appropriate nutritional intake. Aerobic endurance training usually decreases the relative percentage of body fat but has little or no significant effect on fat-free mass. Longer-term programs can result in greater decreases in the percentage of body fat (15, 34, 73). Excessive training may lead to a predominance of catabolic activity in the body and cause an imbalance between catabolic and anabolic processes (141).

Aerobic endurance training results in reduced body fat, increased maximal oxygen uptake, increased exercise economy, increased respiratory capacity, lower blood lactate concentrations at submaximal exercise, increased mitochondrial and capillary densities, and improved enzyme activity.

[Table 6.2](#) lists the physiological changes that occur with short-term (three- to six-month) aerobic endurance training and compares the results for previously untrained and elite aerobic endurance athletes.

External and Individual Factors Influencing Adaptations to Aerobic Endurance Training

A variety of external and individual factors can influence both the acute responses and the chronic adaptations of the cardiovascular and respiratory systems to exercise. The effects of altitude, heat, hyperoxic breathing, smoking, vaping, and blood doping (external factors), as well as genetic potential (individual factor), are briefly described in this section.

Altitude

At elevations greater than 3,900 feet (1,189 m), acute physiological adjustments begin to occur to compensate for the reduced partial pressure of oxygen in the atmosphere (61). [Figure 6.1](#) presents the adjustments to altitude **hypoxia** that are immediate and longer term. Two adjustments that occur early in the acclimatization process are particularly important. First, there is an increase in pulmonary ventilation (**hyperventilation**) at rest and during exercise. This increase in ventilation is primarily the result of increased breathing frequency. With longer stays at high elevation, however, increased tidal volume also contributes to augmented ventilation. Stabilization of ventilation is dependent on the level of altitude and the duration at altitude (77). Second, in the early stages of altitude exposure, cardiac output at rest and during submaximal exercise increases, due primarily to increases in heart rate (61, 108). Submaximal heart rate and cardiac output can increase 30% to 50% above sea-level values, and stroke volume is constant or slightly reduced. Increased submaximal cardiac output reflects the need for increased blood flow at a time when arterial blood oxygen content is reduced in order to maintain adequate oxygen delivery to the tissues.

TABLE 6.2 Physiological Variables in Aerobic Endurance Training

Variable	PREVIOUSLY UNTRAINED SUBJECTS		HIGHLY TRAINED OR ELITE SUBJECTS	
	Before	After	Male	Female
HEART RATE (BEATS/MIN)				
Resting (127, 135, 59)	76.4	57.0	45	51
Maximal (127, 135, 165)	192.8	190.8	196	193.9
STROKE VOLUME (ML)				
Resting (135, 169)	79	76	147	133
Maximal (135, 169, 165)	104	120	187	129.6
CARDIAC OUTPUT (L/MIN)				
Resting (135, 169, 149)	5.7	4.4	8.2	6.7
Maximal (135, 169, 165)	20.0	22.8	33.8	23.5
Heart volume (mL) (127, 135, 149)	860	895	1104	859
BLOOD PRESSURE (MMHG) (127, 135)				
Resting (127, 135)	131/75	144/78	112/75	115/70
Maximal (127, 127a, 135)	204/81	201/74	188/77	180/74
PULMONARY VENTILATION (BODY TEMPERATURE [37 °C] AND PRESSURE [AMBIENT], SATURATED [47 MMHG]) (L/MIN)				
Resting (127, 135)	10.9	12.0	11.8	—
Maximal (56a, 127, 135)	128.7	156.4	164.4	120
ARTERIOVENOUS OXYGEN DIFFERENCE (ML/100 ML)				
Resting (165)	5.8	7.5	—	—
Maximal (135, 169, 165)	16.2	17.1	15.9	17.6

*Data are not from a training study; subjects were untrained or “good distance runners.”
Data compiled by Carwyn Sharp and Kate Baldwin.

Maximal oxygen uptake ($\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) (127, 135, 18, 74)	36.0	48.0	90.6	72.8
% type I fibers (3, 24, 59)	48	51	60.8	60.4-61.8

FIBER AREA (μM^2)

Type I (35)	4,947*	6,284*	6,485	6,069
Type II (35)	5,460*	6,378*	8,342	5,642

CAPILLARY DENSITY

Number/ mm^2 (3, 63)	289	356	640	—
Number/fiber (3)	1.39	1.95	2.15	—

SKELETAL MUSCLE ENZYMES

Citrate synthase ($\mu\text{mol} \cdot \text{min}^{-1} \cdot \text{g}^{-1}$ wet weight) (162)	35.9	45.1	—	—
Lactate dehydrogenase ($\mu\text{mol} \cdot \text{g}^{-1} \cdot \text{min}^{-1}$) (35, 24)	843*	788*	746	744
Succinate dehydrogenase ($\mu\text{mol} \cdot \text{g}^{-1} \cdot \text{min}^{-1}$) (35)	6.4*	17.7*	21.6	10.0
Phosphofructokinase ($\mu\text{mol} \cdot \text{g}^{-1} \cdot \text{min}^{-1}$) (52, 53)	27.13	58.82	20.1	—

MAXIMAL FIBER SHORTENING VELOCITY (FIBER LENGTHS)

Type I (61, 157)	0.99	1.27	1.65	—
Type II (61, 157)	3.18	3.38	3.72	—

*Data are not from a training study; subjects were untrained or “good distance runners.”
Data compiled by Carwyn Sharp and Kate Baldwin.

Within 10 to 14 days at a given altitude, heart rate and cardiac output begin to return to normal values because of the longer-term **acclimatization** response of increased red blood cell production. Thus, with acute exposure to altitude, hyperventilation and increased submaximal cardiac output are rapid and relatively effective responses to offset the challenges of the reduced partial

pressure of oxygen. However, despite these adjustments, during this acute period, arterial oxygen saturation decreases and results in decreases in maximal oxygen uptake and aerobic performance at altitudes above 3,900 feet (1,189 m). Acclimatization changes revert in about one month after return to sea level. As indicated in [figure 6.1](#), chronic physiological and metabolic adjustments that occur during a prolonged altitude exposure include

- increased formation of hemoglobin (generally 5%-15% increase, although higher values have been reported) and red blood cells (30%-50% increase),
- increased diffusing capacity of oxygen through the pulmonary membranes,
- maintenance of the acid–base balance of body fluids by renal excretion of HCO_3^- and through hyperventilation, and
- increased capillarization.

All these adaptations generally improve tolerance of the relative hypoxia at medium and high altitudes (58, 61, 108) and may result in nearly sea-level exercise capacities with adequate acclimatization. A minimum of three to six weeks is needed to adapt at moderate altitude (2,200 to 3,000 ft [608 to 914 m]). However, reduced performance, compared to sea level, is generally expected with altitude exposure regardless of the period of acclimatization. Strength and conditioning professionals are encouraged to inform athletes of both expected acute responses and chronic adaptations to altitude so that they can maintain appropriately adjusted training regimens and a positive mental approach to the impact of altitude.

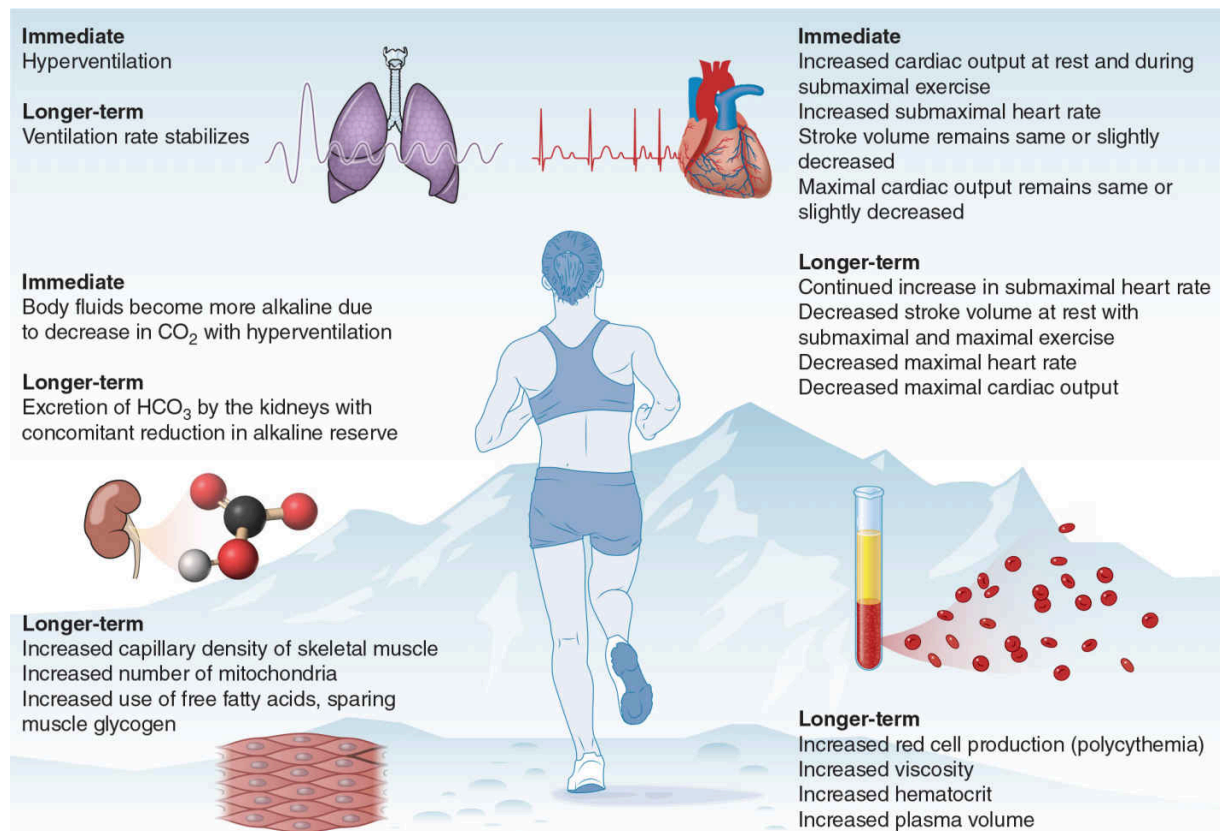


FIGURE 6.1 A summary of the immediate and long-term adjustments to altitude hypoxia.

Heat Acclimatization

The physiologic responses to aerobic exercise depend on the environmental conditions in which an athlete is exercising. Consequently, high environmental temperatures and humidity cause significant alterations to heart rate, oxygen uptake, minute ventilation, and blood lactate responses and, ultimately, performance due to interactions between the environmental conditions and an athlete's thermoregulatory systems (117). If these environmental conditions are hot and dry or warm/hot and humid and produce a sufficient level of heat stress, performance can be impaired due to a cascade of thermoregulatory responses including an increase in sweat rate and blood flow to the skin to facilitate greater heat transfer to the environment. Such a process is in competition with increasing skeletal muscle blood flow to meet the metabolic demands of active muscle (54) and thus restrains blood flow

availability for oxygen and nutrient delivery and metabolic by-product removal in exercising muscle. If this process continues long enough without sufficient fluid intake, dehydration can result, further compounding the issue and putting the athlete under greater physiological and thermoregulatory stress, leading to further compromised performance (54). However, repeated exposure over a long period of time (often referred to as *natural exposure*) to a hot environment can result in thermoregulatory and physiological adaptations referred to as **heat acclimatization**, which can improve an athlete's heat tolerance (10). An athlete can undertake **heat acclimation** protocols (often referred to as *artificial exposure*) to expose the body to such hot environments, thereby raising core and skin temperatures and increase sweating to elicit the same adaptations (124).

The body undergoes a number of well-known chronic thermoregulatory and physiological changes during the heat acclimatization process. These adaptations include an earlier onset of sweating in conjunction with an increase in sweat rate and evaporative cooling, a reduction in resting and exercising skin and core body temperature, an expansion in plasma volume and hence an increase in blood volume, a reduction in resting and exercising heart rate, and a reduction in gastrointestinal distress (49). A summary of the adaptations to thermoregulatory and performance physiology following aerobic exercise in the heat are outlined in [figure 6.2](#).

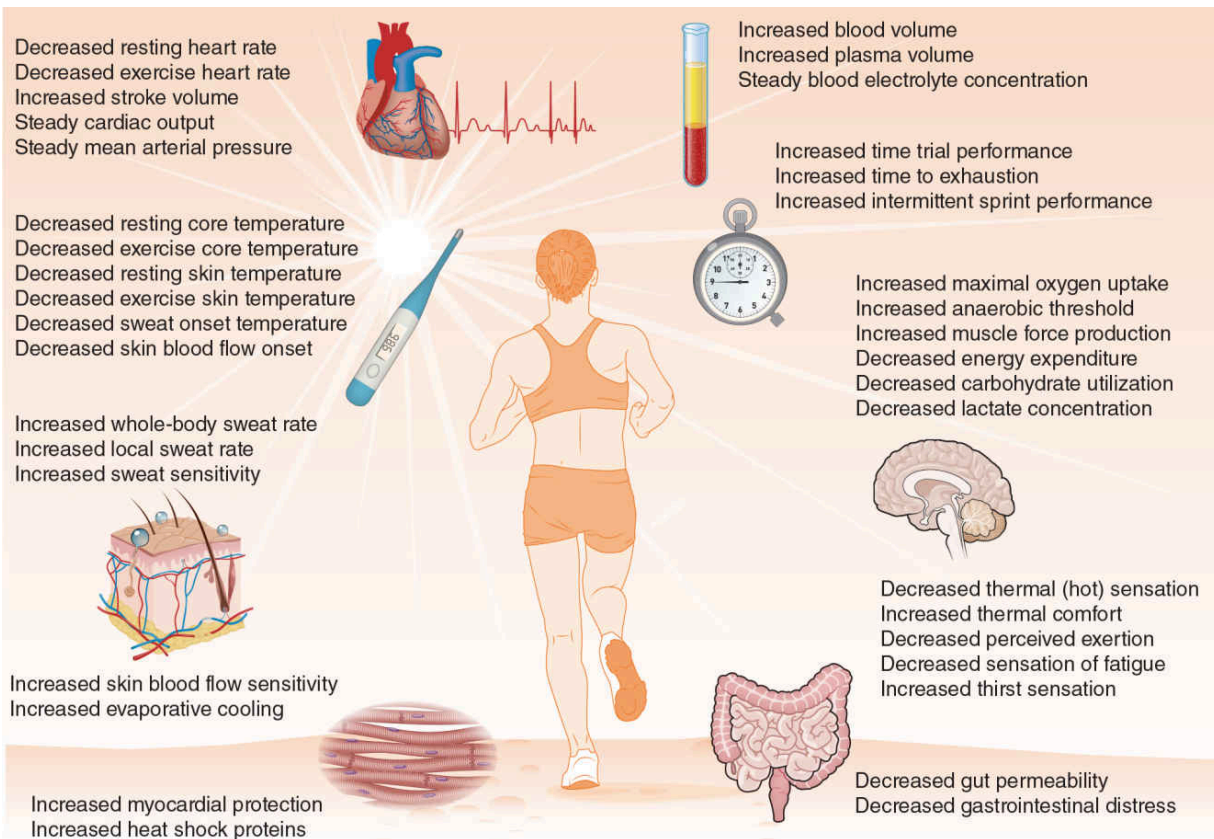


FIGURE 6.2 A summary of the adaptations to thermoregulatory and performance physiology following aerobic exercise in the heat.

Reprinted from O.R. Gibson, C.A. Mee, J.A. Willmott, et al., "Heat Alleviation Strategies for Athletic Performance: A Review and Practitioner Guidelines," *Temperature* 7, no. 1 (2019): 3-36. © 2019 The Authors. Published by Informa UK Limited, trading as Taylor & Francis Group. Distributed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>).

The benefit of these physiological adaptations is that the athlete not only has an improved heat tolerance ability via an improved thermoregulatory system, but such adaptations have also been found to improve time-trial performance (57, 99) and improve maximal oxygen uptake and power output at lactate threshold (99). Additionally, the athlete's perception of effort and level of thermal comfort have also been found to be improved following heat acclimation (49). However, in a review by Rahimi and colleagues (130) of the physiological responses to heat acclimation across 11 randomized controlled trials with 215 participants, it was reported that heat acclimation was effective for improving time-trial

performance as well as reducing time-trial heart rate and maximal heart rate, despite there being no beneficial effect on maximal oxygen uptake, core or mean skin temperature, plasma volume, or blood lactate, nor meaningful effects on rating of perceived exertion or thermal comfort. The differences between studies and whether improvements were observed with respect to heat acclimation may be related to the environmental conditions (i.e., warm or hot, dry or humid), intensity of the exercise used, the acclimation protocols that the athletes were exposed to, and the initial level of athlete acclimation, all of which determine the degree of thermoregulatory and physiological adaptations (124).

Heat acclimation adaptations start to occur quickly once exposed to the hot environment and continue for the duration of the process. However, it needs to be noted that heat acclimation adaptations do not all occur at the same rate, with some occurring more quickly than others and some (e.g., sweat rate) continuing as the acclimation period continues (124). The time course of the physiological adaptations to heat acclimation are outlined in [figure 6.3](#). The physiological changes to the thermoregulatory system start to occur very quickly, commencing from the first day of exposure (124) and continuing over subsequent days, with the majority of the adaptations (~75%) reported to occur within the first four to six days (121).

Unfortunately, heat acclimation-induced thermoregulatory and physiological adaptations are transient and relatively short-term, and after leaving the hot environment the changes start to dissipate after only a few days or weeks depending on the length of the exposure (47). While a lot is known about the heat acclimation process, considerably less is known about the **decay rate** of the adaptations (49). Originally, it was proposed that for each day of a heat acclimation protocol, the adaptation would be retained for two days (96). However, more recent research indicates that for every day that passes without exposure to the heat, there is a 2.5% loss of adaptations (28). The adaptations also decay at different rates and

coincide with the rate of adaptation; that is, those that adapted quickly also decay quickly (122). Typically, full heat acclimation adaptations are expected to be lost after a month if there is no repeated exercise heat stress (128). It also appears that the loss of adaptations may be affected by the level of exercise undertaken after heat acclimation. Consequently, it is recommended that athletes maintain their training in the days after heat acclimation to assist retaining their thermoregulatory and physiological adaptations (128).

It has also been reported that the heat acclimation adaptations may remain longer for those athletes acclimatized in dry, hot environments compared to humid, hot environments (121). Unfortunately, little research has been conducted on the transferability of adaptations between different hot environments (e.g., hot and dry to warm and humid). However, in one of the few studies to investigate this aspect of heat acclimation, it was found that cross-adaptation from one environment to the other is achievable (56), an important point worth noting for athletes who train and compete in different hot environments.

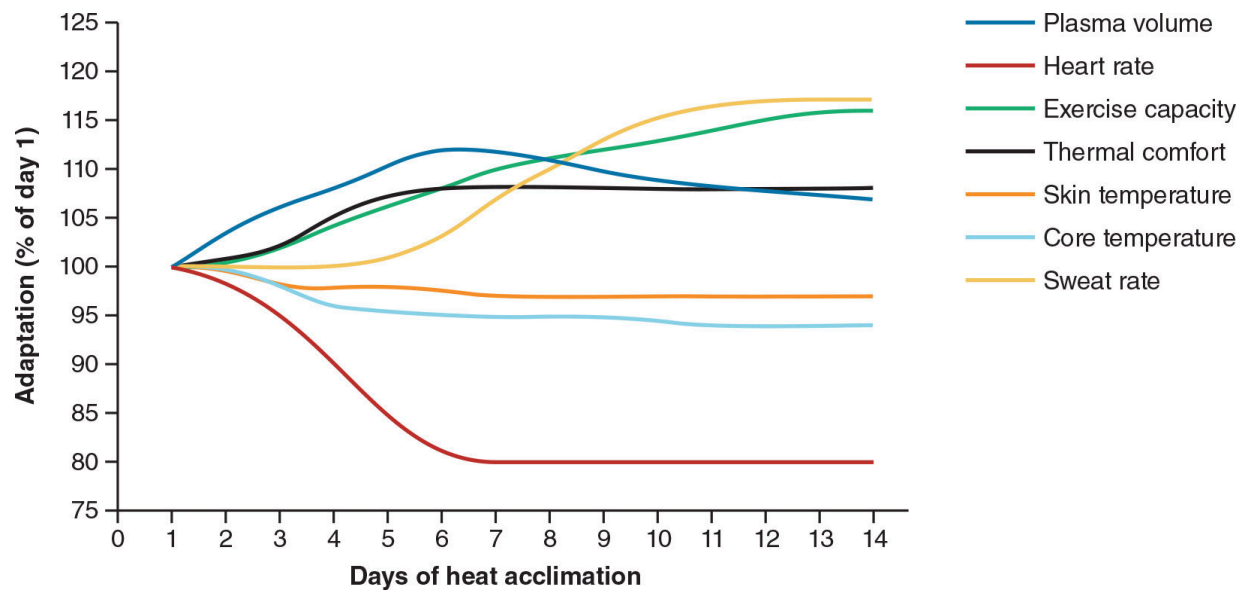


FIGURE 6.3 The time course of the physiological adaptations to heat acclimation.

Reprinted by permission from J.D. Périard, S. Racinais, and M.N. Sawka, "Adaptations and Mechanisms of Human Heat Acclimation: Applications for Competitive Athletes and Sports," *Scandinavian Journal of Medicine & Science in Sports*, 25 (2015).

Two other factors to consider with respect to heat acclimation are the influence of the level of aerobic fitness and biological sex. There is evidence to suggest that athletes who have a higher level of aerobic fitness have a greater ability to exercise in the heat due to the ability to tolerate higher core body temperatures (124) and may also better retain their heat acclimation adaptations (121). With respect to sex, differences are observed in the physiological responses to exercise in the heat between males and females. Females have been found to experience greater increases in body temperature in the heat due to a lower total body sweat response when compared to males (43). This difference in sweat rate is attributed to a smaller increase in sweat rate for a given increase in body temperature (i.e., lower thermosensitivity) (43). It has also been observed that while both females and males respond to short-term heat acclimation protocols, females may require a slightly longer protocol to establish thermoregulatory and cardiovascular stability (111).

With this information in mind, strength and conditioning professionals need to consider the type of heat acclimation protocol that they will employ, the proximity of the heat acclimation protocol to the athlete's competition, and the training status and sex of the athlete with whom they are working, in order to maximize the potential benefits from the acclimation environment to the competition environment. More information on heat acclimation protocols and the different types and considerations when implementing such protocols are outlined in [chapter 21](#).

Altitude or Heat: Which Training Protocol Should an Athlete Undertake?

Before answering this question, it is important to understand how the two types of training protocols differ in terms of adaptations. Heat acclimation training has become popular with aerobic endurance athletes, often as an alternative to altitude training due to less expense and easier access to hot versus altitude training environments. While evidence suggests that heat acclimation can improve maximal oxygen uptake in **thermoneutral** (164) and hot environments (99, 163) as well as performance in thermoneutral (112) and hot environments (57, 99), the changes are not due to the same physiological mechanisms brought about by altitude training (hypoxic exposure).

Training at altitude causes exposure to a hypobaric-hypoxic environment and a reduction in an athlete's arterial oxygen content, which sets off a sequence of events starting with an increase in the amount of hypoxia-inducible factor and the subsequent expression of the genes associated with the transcription of **erythropoietin (EPO)** and nitric oxide (6). The increased concentrations of EPO bring about changes in red cell volume and hemoglobin mass, which increase the oxygen-carrying capacity of the blood as well as increase the arteriovenous oxygen difference, which can provide an improvement in maximal aerobic capacity and subsequent aerobic endurance performance (6). In comparison, heat acclimation training

targets changes in the thermoregulatory system via increases in core body temperature, which in turn brings about a reduction in central venous pressure and renal perfusion and subsequent increase in aldosterone secretion (6). The result of this process is a sudden and large expansion in plasma volume, which in turn increases stroke volume and hence is proposed to improve maximal aerobic capacity and subsequent aerobic endurance performance (6). A graphical representation of the proposed physiological mechanisms of altitude training and heat acclimation training and their subsequent effect on aerobic endurance exercise performance is provided in [figure 6.4](#).

In view of the potential benefits to both altitude and heat training as outlined above, one might think that completing both forms of training together would provide an athlete with the best of both worlds—that is, **erythropoiesis** from hypoxia stimulation and plasma volume expansion from heat acclimation. Unfortunately, the answer is not that simple, for altitude training can evoke an acute reduction in plasma volume, which is in contrast with the expansion of plasma volume experienced with heat acclimation (118). In addition, the time course of adaptations to altitude (hypoxic exposure) and heat acclimation appears to be different, with hypoxia exposure generally requiring slightly longer exposure times compared to heat acclimation (60).

Research has been conducted to investigate the efficacy of combining the two forms of training (i.e., heat acclimation with hypoxic exposure) on performance in a hot environment using trained runners (109). Even though heat training combined with living-high/training-low hypoxia produced a greater physiological stress, the thermoregulatory adaptations for the combined group compared to an independent heat training group were limited and found to provide no additional physiological benefit when exercising in a hot environment (109). A similar study by Hanson and colleagues (60) comparing a heat acclimation-alone group with a group combining heat and hypoxia exposure on cycling endurance performance in a hot environment observed similar improvements in

performance for short-duration exercise across the groups. The results of these studies suggest that even though the physiological adaptations to altitude and heat training may be different, hypoxic exposure does not provide any additional benefit above and beyond heat acclimation as it relates to exercise performance in the heat, nor does it impair heat acclimation.

In light of these research findings, which training protocol (altitude or heat) should an athlete complete? In answering this question, it is important to remember two points. First, nonhematological adaptations are directly related to the stresses of the training environment, and second, the nonhematological adaptations to both altitude and heat are specific to the environment in which the athlete will compete (118). Or put another way, what is going to have the biggest impact on the athlete's performance on race day, altitude or heat? For example, if an athlete was preparing for a marathon to be held at an altitude of 2,500 to 3,000 meters (8,202-9,843 ft), the answer would be altitude, because the athlete would notice the change in atmospheric pressure at that altitude if he or she were from a sea-level environment. However, if the athlete was preparing for a marathon at sea level but in a warm to hot and humid environment and notably above his or her temperate, dry training environment, he or she would be best served by undertaking a heat acclimation protocol. Consequently, it is necessary to compare the training and competition environments and consider the athlete's current level of adaptations to determine what adaptations should be optimized to improve performance on race day. Irrespective of the training method selected to prepare an athlete for competition, establishing how each athlete adapts and providing individualized programs to optimize adaptations is important, because not all athletes will respond to altitude and heat protocols the same (118).

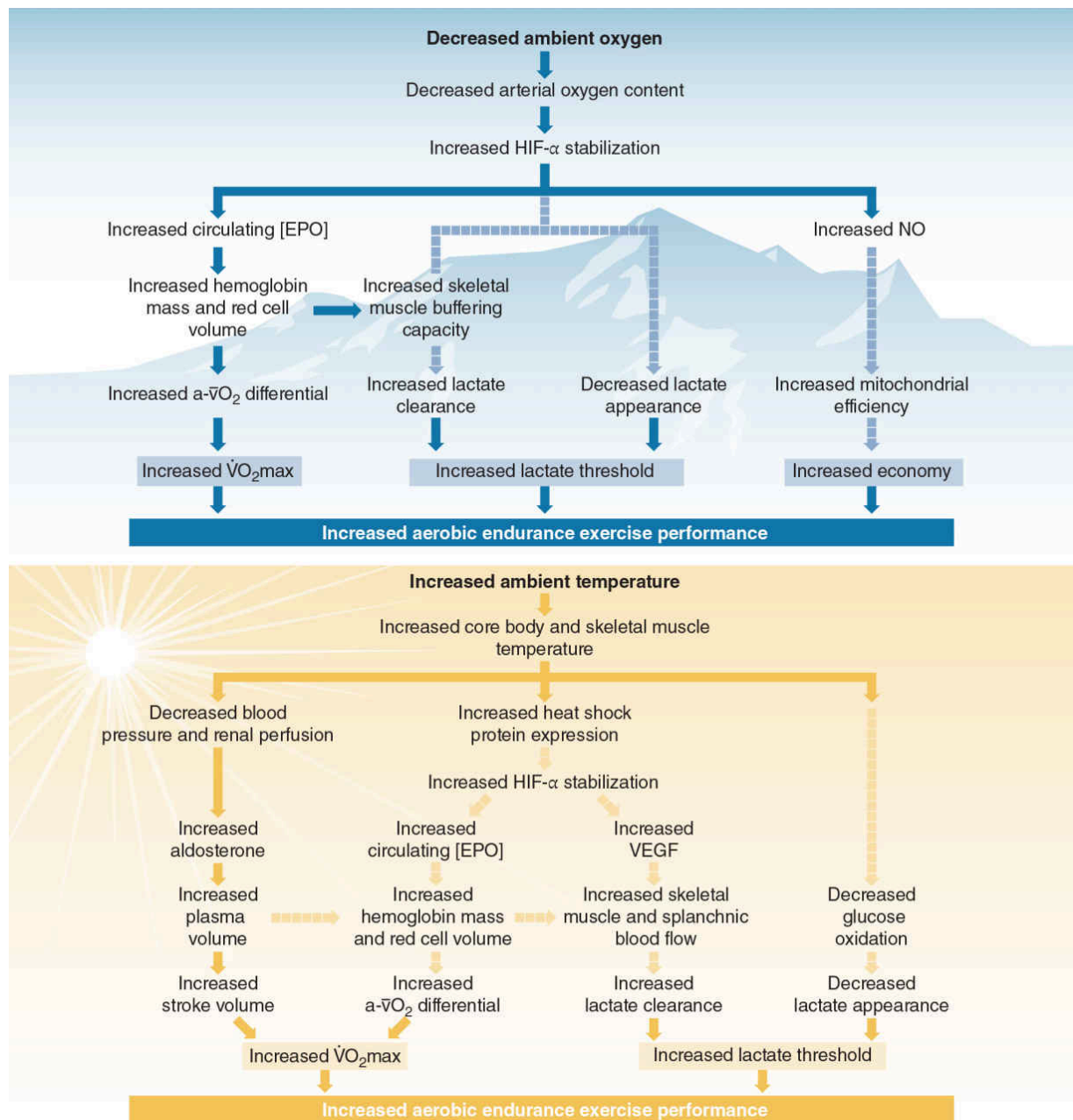


FIGURE 6.4 A graphical representation of the proposed physiological mechanisms of altitude and heat acclimation training and their subsequent effect on aerobic endurance exercise performance. Erythropoietin = EPO; nitric oxide = NO; red cell volume = RCV; vascular endothelial growth factor = VEGF; hypoxia-inducible factor alpha = HIF- α ; maximal aerobic capacity = $\dot{V}O_{2max}$; arteriovenous oxygen difference = a- $\dot{V}O_2$ diff. Solid lines indicate well-established relationship and dashed lines indicate limited evidence supporting the relationship.

Adapted by permission from M.N. Baranauskas, K. Constantini, H.L. Paris, et al., "Heat Versus Altitude Training for Endurance Performance at Sea Level," *Exercise and Sport Sciences Reviews* 49, no. 1 (2021): 50-58.

Hyperoxic Breathing

Breathing oxygen-enriched gas mixtures (**hyperoxic breathing**) during rest periods or following exercise can result in short-term improvements in oxygen transport, lactate metabolism, power output, and work tolerance when compared to **normoxic breathing** (21). In contrast, studies examining the systematic supplementation of hyperoxic breathing during aerobic endurance training over time show contradictory results and overall small benefits (21). Hyperoxic breathing has been proposed to increase the amount of oxygen carried by the blood and therefore increase the supply of oxygen to working muscles and reduce central fatigue (21). However, the breathing of ambient air at sea level by healthy people results in arterial hemoglobin oxygen saturation between 95% to 98%. Thus, the potential for hyperoxic breathing during rest periods or following exercise has yet to be fully elucidated (58, 126).

Smoking and Vaping

Relatively little research is available regarding smoking and exercise performance, possibly because athletes and active individuals may tend to avoid smoking for fear of impairing performance or increasing disease risk (100, 123). However, based on the available evidence, it is clear that smokers experience lower oxygen uptake, impaired heart rate responses to exercise, and a decrease in muscular endurance compared with nonsmokers (9, 91). Smokers experience impairments in lung function and are at increased risk for chronic obstructive pulmonary diseases (104, 148) including chronic bronchitis (79) and emphysema (101). Decrements in exercise tolerance and cardiopulmonary function have been observed in teenagers (100), demonstrating that the harmful effects of smoking occur in even young smokers within a relatively short period of time. The detrimental effects of smoking include

- increased airway resistance due to nicotine-related bronchiole constriction or increased fluid secretion and swelling in the

bronchial tree due to the irritation of smoke; and

- paralysis of the cilia on the surfaces of the respiratory tract by nicotine, which limits the ability to remove excess fluids and foreign particles, causing debris to accumulate in the respiratory passageways and adding to the difficulty of breathing.

Thus, even the light smoker may feel respiratory strain during exercise and experience a reduction in the level of performance (58, 108).

Carbon monoxide, a component of cigarette smoke, is associated with an impaired hemodynamic response to exercise and increased catecholamine release. Carbon monoxide has a higher affinity for hemoglobin than does oxygen. The resulting **carboxyhemoglobin** (carbon monoxide and hemoglobin) reduces the amount of oxygen that can be carried by hemoglobin and thus reduces the oxygen that can be provided to the working muscles. The reduction in oxygen-carrying capacity may reduce maximal exercise capacity, and submaximal cardiovascular responses may increase in an effort to provide adequate oxygenated blood to the working muscles. The increased catecholamine release increases heart rate and blood pressure.

Vaping refers to the act of inhaling an aerosol often containing nicotine and flavoring through a device, commonly known as an *e-cigarette* or *vape pen*, which heats a liquid to generate vapor without the combustion associated with traditional cigarette smoking. Vaping is currently the most common modality of nicotine consumption in young people, with 27.5% of high school students in 2019 in the United States reportedly vaping (133). The toxicants found in vapors are reportedly lower in quantity when compared to traditional cigarette smoking; however, adverse short-term side effects on the respiratory system have been reported. Research examining the chronic effects of vaping on lung health and long-term exercise performance is extremely limited due to the relatively new widespread uptake of vaping. Currently, no large-scale

prospective cohort study has examined the link between vaping and long-term adverse respiratory outcomes. This is further complicated by the large spectrum of vaping options including various flavors, additives, and nicotine concentrations. However, in vaping products that contain nicotine (over 99% of products), the nicotine-stimulated catecholamine release can activate receptors in the heart and result in an increased heart rate, blood pressure, and cardiac contractility and workload and, in the long term, can result in changes in cardiovascular physiology and the development of heart failure and other associated chronic health issues (133).

Studies examining short-term side effects of vaping report inflammation and cellular damage in the lungs, leading to decreased lung capacity and impaired gas exchange (22, 151), which may in turn affect performance and recovery times during exercise. However, whether these changes are maintained over the long term and after vaping cessation is still yet to be established. Studies also demonstrate that both acute and subacute vaping can exacerbate both asthma and chronic obstructive pulmonary disease and induce physiological changes in the airways and immune system (159), which can negatively affect performance and recovery in athletes. Very few studies have examined the effect of vaping directly on aerobic endurance exercise. In young, healthy adults, regular vaping use significantly reduced cardiorespiratory fitness, measured by $\dot{V}O_{2\text{peak}}$ when compared to nonvapers of the same demographic (142a). The young adults also demonstrated lower chronotropic responses and reduced O_2 extraction and use during exercises compared to nonusers. In addition, significantly slower 2-mile (3.1 km) run times have been reported in vape users compared to nonvapers, with times also being slower compared to traditional cigarette smokers (30). Additionally, vaping can cause short-term increases in heart rate and diastolic blood pressure primarily by the systemic release of catecholamines (144).

Blood Doping

The practice of artificially increasing red blood cell mass (**blood doping**) as a means to improve athletic performance has been criticized as unethical and as one that poses serious health risks to the athlete (120). Nonetheless, research has suggested that this practice can improve aerobic exercise performance and may enhance tolerance of certain environmental conditions (138).

Blood doping can be accomplished through infusion of an individual's own red blood cells or those from someone else, or through administration of EPO, which stimulates red blood cell production. Infusion of red blood cells rapidly increases red blood cell mass but for just a few weeks (116), whereas EPO takes longer to cause an increase in red blood cell production but the effect lasts as long as EPO is administered (138). In either case, it is theorized that increasing red blood cell mass increases the blood's ability to carry oxygen and thus increases oxygen availability to working muscles. Maximal oxygen uptake has been demonstrated to increase by up to 11% with either red blood cell infusion or EPO administration (138, 139). At standardized submaximal workloads, blood doping has also been associated with decreased heart rate and blood lactate as well as higher pH values (138).

The effects of blood doping suggest the potential for diminishing the impact of various environmental conditions. The effects of altitude appear to be lessened with blood doping, although as altitude increases, the positive influence of blood doping is reduced (132). Environmental stressors such as heat and cold exposure may also be affected by blood doping. During heat exposure, a blood-doped athlete will be able to tolerate submaximal exercise stress more easily (76, 137-139). The increased blood volume associated with increased red blood cell mass allows the body to shunt more blood to the skin for improved thermoregulation while still providing sufficient blood to transport oxygen to the working muscles. However, blood doping appears to confer these benefits primarily to individuals who are already acclimatized to heat and provides very

little aid relating to the response to heat of those who are not acclimatized (138). Much less is known about blood doping with respect to cold stress, and although some theoretical benefit has been suggested, this practice could conceivably increase health risks (138).

The health risks associated with blood doping further complicate the controversy surrounding this practice. Theoretically, high hematocrit levels may increase risks for embolic events such as stroke, myocardial infarction, deep vein thrombosis, or pulmonary embolism. Increased arterial blood pressure, flu-like symptoms, and increased plasma potassium levels may present themselves with EPO use (139). Finally, there is risk associated with infusion or transfusion, although in most cases the risk is relatively small (139).

Genetic Potential

The upper limit of an individual's genetic potential contributes significantly to the absolute magnitude of the training adaptations. The magnitude of change also depends on the current training status of the individual. Each biological system adaptation, such as that of the cardiovascular system, has an upper limit, and as the athlete gets closer to that upper limit, smaller and smaller gains are observed. For example, in some elite competitions (e.g., swimming), small gains in performance may be the difference between the gold medal and 26th place. Thus, in an event in which tenths or hundredths of a second make a big difference, it may be worth the extra training time to gain that 0.05% improvement in performance. Because of the small gains possible with training in elite athletes, careful program design and monitoring become even more critical (34, 83, 89, 168).

Detraining

Detraining is defined as the partial or complete loss of training-induced adaptations in response to an insufficient training stimulus (114, 115). Detraining is governed by the principle of training reversibility, which states that whereas physical training results in several physiological adaptations that enhance athletic performance, stopping or markedly reducing training induces a partial or complete reversal of these adaptations, compromising athletic performance. A distinction needs to be made with regard to cessation of training that results in a reversal of adaptations and tapering. **Tapering** is the planned reduction of volume of training (usually in duration and frequency but not intensity) that occurs before an athletic competition or a planned recovery microcycle. This type of reduction in training is designed to enhance athletic performance and adaptations.

Aerobic endurance adaptations are most sensitive to periods of inactivity because of their enzymatic basis. The exact cellular mechanisms that dictate detraining changes are unknown, and further research is needed to clarify the underlying physiological alterations. Two review articles (114, 115) address the factors that contribute to aerobic performance, most importantly $\dot{V}O_{2\max}$ and the factors determining $\dot{V}O_{2\max}$ (cardiac output and arteriovenous oxygen difference) and the impact of detraining. The authors discuss the impact of short-term (four weeks) (114) and long-term (more than four weeks) (115) detraining. In highly trained athletes, maximal oxygen uptake is reduced 4% to 14% with short-term (27, 106) and 6% to 20% with long-term detraining (26, 33, 106). The reduction in $\dot{V}O_{2\max}$ is primarily a result of decreased blood volume (25), decreased stroke volume (25, 26), decreased maximal cardiac output (25, 26), and increased submaximal heart rate (25, 31, 33, 105). The decrease in $\dot{V}O_{2\max}$ supports the loss of aerobic endurance performance (25, 26, 31, 33, 72, 105).

Proper exercise variation, intensity, maintenance programs, and active recovery periods can adequately protect against serious detraining effects (36).

Conclusion

Aerobic endurance exercise produces many acute cardiovascular and respiratory responses, and aerobic endurance training produces many chronic adaptations. This information can be of particular value for developing the goals of a conditioning program and can provide a basis for clinical evaluation and the selection of parameters to be included in such an evaluation process. Knowledge of cardiovascular, respiratory, nervous, muscular, bone and connective tissue, and endocrine system responses to aerobic endurance training can help the strength and conditioning professional understand the scientific basis for aerobic conditioning and the adaptations to expect and monitor during training. Adaptations to specific types of exercise stimuli take place in the body. Optimal adaptations reflect careful design, implementation, and performance of strength and conditioning programs.

KEY TERMS

acclimatization
blood doping
bradycardia
carboxyhemoglobin
concurrent training
critical power
decay rate
detraining
erythropoiesis
erythropoietin (EPO)
exercise economy
heat acclimation
heat acclimatization
hemoglobin
hybrid fiber

hyperoxic breathing
hyperventilation
hypoxia
lactate threshold
mitochondria
myoglobin
normoxic breathing
onset of blood lactate accumulation (OBLA)
repeated sprint-sequence interval session
tapering
thermoneutral
vaping

STUDY QUESTIONS

1. Which of the following is not a physiological adaptation to aerobic endurance training?
 - a. stroke volume increases
 - b. submaximal heart rate decreases
 - c. exercise economy increases
 - d. fat oxidation decreases
2. During the heat acclimatization process, which of the following chronic physiological thermoregulatory and performance changes does an athlete not experience?
 - a. increase in plasma volume
 - b. reduced lactate concentration
 - c. decrease in evaporative cooling
 - d. decrease in exercise core temperature
3. Which of the following statements is not true concerning altitude and heat training?
 - a. The time course of adaptations to altitude and heat acclimation appears to be different, with altitude training generally requiring slightly longer exposure times compared to heat acclimation.

- b. Heat environments cause exposure to a hypoxic environment and a reduction in an athlete's arterial oxygen content.
 - c. Heat and altitude training adaptations occur in response to different physiological mechanisms.
 - d. Heat acclimation training targets changes in the thermoregulatory system via increases in core body temperature.
4. Which of the following short-term effects is seen after vaping?
- a. increases in heart rate and diastolic blood pressure
 - b. decreases in heart rate and systolic blood pressure
 - c. an improvement in systemic inflammation
 - d. there are no short-term negative effects from vaping
5. Which of the following is not a benefit seen from hyperoxic breathing?
- a. short-term improvements in oxygen transport
 - b. short-term improvements in lactate metabolism
 - c. short-term improvements in power output
 - d. short-term improvements in exercise economy